

Laboratory rules for the team projects
in labs 0.74, 0.75 and 0.105 of building D11, including the AutFab,
of the Darmstadt University of Applied Sciences

1) General rules:

- a) Unauthorized persons are prohibited from entering the laboratory.
- b) Contact with open terminals or any live part is prohibited.
- c) The instructions and commands of the supervisors must be followed.
- d) Eating and drinking (except water) is not permitted in the laboratory.
- e) Smoking and noise (e.g. with radio, PC speakers, smartphone speakers, etc.) are not permitted in the laboratory.
- f) Every student working in the laboratory must read, sign and comply with these laboratory regulations and read and comply with the fire safety regulations.
- g) Persons who violate the laboratory regulations may be expelled from the laboratory and lose admission to the team project.

2) Attention:

Since during laboratory tests a danger to the testing persons from electrical voltages cannot be excluded, the persons present must protect themselves from accidents by taking special care.

Take care when sitting on wheelchairs. Do not tilt the wheelchair!

3) Stay:

Students may only be in the laboratory if at least two persons are present.

4) Safety measures:

- a) For some measurements, the housing (e.g. of an oscilloscope) must be disconnected from the protective conductor. In this case, special care must be taken with metal housings.
- b) In general, the following safety measures apply:
 - Experimental set-ups where operating voltages higher than 42V may occur may only be operated at experimental stations, which are protected by residual current circuit breakers and emergency switch off (German: Not-Aus) buttons. It may only be switched on if at least 2 persons are present in the laboratory.

- In case of any accident or injury call a first-aider (German: Ersthelfer) of the faculty, e.g.: Mr. F. Radler, Mr. C. Schulz, Mr. G. Trautmann, Mr. A. König, Mr. H. Weinerth, Mr. M. Anthes, or a person indicated as first-aider on the emergency guide (GERMAN: Notfallplan) at the first-aid kit.
- Persons who have come into contact with voltages above 50 V_{rms} must be immediately examined by an emergency physician (German: Notarzt).

Attention: In case of an emergency call, request EMERGENCY AMBULANCE (German: Notarztwagen)!

- In the corridor outside the exit of the laboratory 0.75 (AutFab), there are a first-aid kit (German: Erste-Hilfe Kasten) and a defibrillator hanging on the wall between the laboratory of automation systems and the laboratory of electrical measurements, which can be used if necessary. Please follow the instructions posted on/near the kit or the defibrillator.
- In case of respiratory arrest and/or cardiac arrest, artificial respiration and, if necessary, cardiac massage must be performed immediately. These measures must be carried out until a doctor has arrived.

4.1 Special instructions when working with the Smart Factory AutFab

When working on the Smart Factory AutFab, in addition to the previously described general dangers due to electrical voltage, a danger of injury due to dangerous movements cannot be excluded. Further precautions must therefore be taken:

- a) Red emergency stop (German: Not-Halt) buttons are fitted around the AutFab. When actuated, the dangerous movements are stopped throughout the AutFab. However, the system itself, the lighting and the wall sockets remain live. The main power supply of the whole system can be switched off at the main control station Station 10 by means of the main switch.
- b) The power supplies of the AutFab are connected to the main switch via residual current circuit breakers, which switches off at a fault current of approx. 30 mA.
- c) The control cabinets installed in the system (at the control station and at the high-bay warehouse) must not be opened when the main power supply is switched on, or at the high-bay warehouse when the power supply to the control cabinet of the high-bay warehouse is switched on!
Attention, even if the power supply is switched off, the components in the switch cabinets may still have a dangerous amount of electrical energy stored! Do not touch any components under any circumstances!
- d) Before working on central electrical parts of the AutFab, the system must be switched off via the main power supply.
- e) Before working on electrical parts of individual stations, the area stop (GERMAN: Bereichshalt) of the station must be actuated and at least the station must be disconnected from the power supply (at least by means of the rotary switch in the control section of the station). Prevent reconnection. Afterwards, test for the absence of harmful voltages before starting work. If necessary, insulate adjacent

live parts.

Before working on mechanical parts of the individual stations, at least the station has to be switched to stop mode and the area stop button has to be pressed.

- f) The existing safety systems must not be bypassed, manipulated, deactivated or changed.
- g) It is strictly forbidden to enter with the full body or with any limbs enclosed areas when the system is switched on. When the system is switched off/secured, access to the fenced areas is only permitted if a supervisor is nearby and can be reached at short notice.
In case of any problems or potential destruction, do not reach into the enclosed areas! No component is worth risking your health.
- h) Ensure that the shuttle can travel freely, also with regard to the side cheeks of the shuttles. Do not touch the moving shuttle or between the shuttle and the transport rail. Do not touch the area of the monorail switch (turnout)!
- i) Do not look directly into the light signals of the system (e.g. of the photoelectric sensors, laser scanner, infrared sensors of the shuttles at the front, etc.)! If you accidentally look into the light signals, look into another direction as quickly as possible. In case of an emergency, ask any staff or student nearby to call the emergency ambulance.
- j) Do not hold on to the metal superstructures/metal sections, do not lean against them, do not tear on them and do not sit on parts of the AutFab or on the tables.
- k) Students may operate the "Manual operation mode" or the "Setup mode/Teaching mode" unsupervised after a detailed briefing, but only if there is nobody in the fenced area. At least two persons must be present.
- l) When leaving the AutFab (even for a short time), it must be secured, i.e. the central emergency stop must be triggered by pressing the central emergency stop button.
- m) After the end of the work on the AutFab, the system must be shut down and switched off according to the instructions given by the supervisors.

5) Conducting the courses / projects.

- a) The equipment and tools used must be handled with care. Damage to the devices and accessories must be reported immediately. The person responsible is liable for damages caused by gross negligence.
- b) If the device is rented, replacement must be provided for lost items.
- c) When setting up the circuits, all test devices must be set up in a sensible manner, voltage sources must be protected (concerning contact, short circuit, etc.), first main circuits, then secondary circuits must be switched in a clear manner and all measuring, control and regulating devices must be set to an uncritical mode before switching on.

When disconnecting the mains plug of frequency converters, do not touch the contacts of the plugs, as they are still live for a while due to the energy stored in the converter!

- d) In the case of mechanical structures, e.g. drives, the dangerous movements must be protected. This protection must be shown to the supervisor and approved by him before the test setups are used.
- e) If a group of students has completed an electrical or mechanical test setup, the supervisor must inspect the setup and approve it for operation. If the assembly is changed, the voltage must be switched off beforehand.
- f) In the case of minor changes, the students can switch on again independently in consultation with the supervisor.
- g) If a special danger is detected, one of the red emergency stop buttons must be pressed immediately, in order to stop the dangerous movements. Please note, however, that this interrupts the entire operation of the system for a certain time, until the system has been restarted. Intentional activation may lead to an exclusion of the project.
- h) Before dismantling a test circuit, all switches and circuit breakers must be switched off and the voltage source must be disconnected from the installation at first.
- i) At the end of the daily work on the AutFab or on the test rig:
 - Shut down the computers of the test setup that are no longer needed.
 - If no one else is working on the AutFab, shut down all computers that have not yet been shut down and then switch off the system according to the instructions, given by the supervisors.
 - If you have worked in the project booth of lab 0.74 and no one else is working there anymore, shut down all computers in that area. If also no one works in the booth of the other side of lab 0.74 anymore turn off the electricity of lab 0.74 by the main switch at the wall next to the door.
 - If used, return all parts such as HoloLens, adapters, test leads, etc. to the component cabinet and lock it.
 - If no one is working in the AutFab area or in the test setup area (project area of lab 0.74) or in the whole lab 0.74 anymore, switch off all lights in this area, or if no one is working in the entire laboratory hall anymore, switch off all lights in the laboratory hall.
 - If nobody is working in the AutFab area any more, the door of the laboratory 0.75 must be locked from the outside, when leaving the laboratory. If the experiment is in laboratory 0.74 and nobody is working in this area any more, the door of laboratory 0.74 must be locked from the outside, when leaving the laboratory. If no one is working in the entire laboratory hall any more, both doors of the laboratory hall (0.74 and 0.75) must be locked from the outside, when leaving the laboratory.
 - All doors in the laboratory are equipped with fire protection systems. This means that they can be opened from the inside at any time. However, if they were previously locked, they cannot subsequently be opened from the outside without a transponder. If students leave the laboratory temporarily, e.g. to go to the toilet,

they should therefore ensure that they take the transponder with them, so that they can return into the laboratory.

6) Non-disclosure and software

- a) The software on the computers or possibly provided software may not be copied and (even if the software is free of charge) not passed on, unless this was or is expressly permitted (public domain, free goods or conditional shareware).
- b) Any passwords used must be kept secret. The user, as well as the supervisor are to be informed immediately about the knowledge of foreign passwords or if it has been determined that unauthorized persons have gained knowledge of foreign passwords.
- c) Software, which is issued or installed on the project/AutFab PCs may only be used for educational or research purposes within the university. It is also subject to the prohibition of disclosure (see 6 a)) as well as the general legal and the special, company-dependent license conditions.
- d) It is strictly forbidden to download, copy, delete, move or destroy licenses or software located on the computers.
- e) The use of computers in the network is limited to the computers in the laboratory.
- f) If the student gains insight into the structure and content of materials protected by companies or experience with components protected by companies, the student may only use this knowledge for the project being worked on and may not pass it on to third parties, including other students who are not working on the same project. This knowledge must be kept secret during the project and also after completion of the project. This applies in particular to all data, documents and knowledge that the student has acquired in the course of projects with project partner companies, pilot projects, incubation projects, or similar projects in written or oral form from companies or from the university supervisors. The student is not authorized to reproduce, construct, imitate or manufacture software, products or other objects in whole or in part.
- g) The student agrees that the university may use and exploit the knowledge gained and information transmitted in the project free of charge and without restriction.
- h) If the project is a pilot, incubation project or similar in cooperation with a company, the student agrees that the partner company may use and exploit the knowledge gained and information provided in the project free of charge and without restriction.
- i) Exceptions:
 - Software created by the students for the respective experiment may only be copied by the respective owner, provided no third party has any interests in it.
 - Further exceptions can only be made in writing by one of the supervising professors or the responsible software representative of the department.

- j) The usage of USB storage media or storage of data in cloud systems is only permitted after consultation with a supervisor.
- k) The installation of software on the laboratory computers is only permitted after consultation with a supervisor.

I hereby declare that I have read and understood the fire safety regulations ("BrandschutzInformation Studierende") and the above laboratory regulations for working in the laboratories 0.74 & 0.75 in building D11, including the AutFab, and that I will comply with these regulations and follow the instructions and commands of supervisors and posted instructions.

Surname, First Name

Place,

Date,

Signature